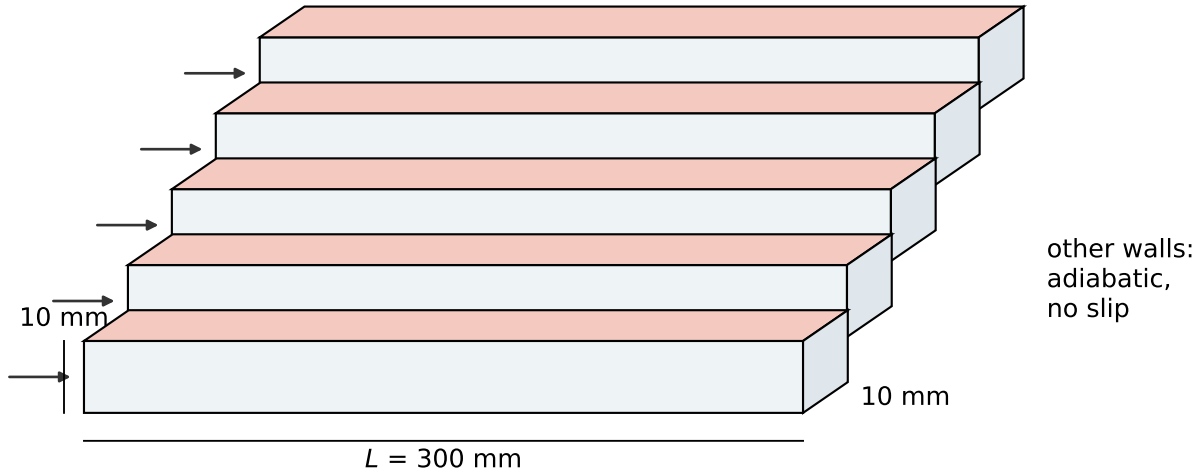


(a) cold-plate heat-exchanger domain

heated wall (top of every duct): uniform $q_w = \dot{Q}/A_{heated} = 627 \text{ W m}^{-2}$

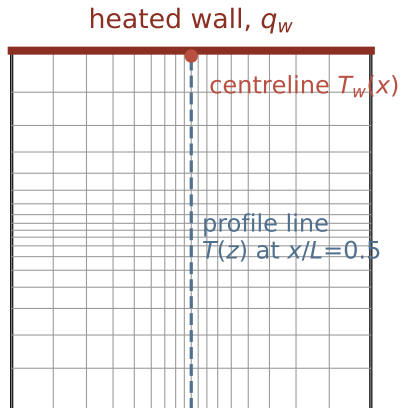


inlet: $T_{in}, \dot{m}$ (He: mass-flow BC; LN₂: velocity BC)

outlet: fixed p

five parallel square ducts, $D_h = 10 \text{ mm}$, pitch 18 mm ; total heated area 0.015 m^2 , flow area $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct
two-sided grading, ratio 6
first cell 0.19 mm
180 cells along L
 2.9×10^5 cells in total